

UNIVERSITY OF CALIFORNIA, SAN DIEGO

CONSENT TO PARTICIPATE IN RESEARCH

1. Study Title and Number

Title: Mechanisms of Language and cognitive development in infants, children, and adults.

Study Number: 811233

2. Principal Investigator

Martin Zettersten, Ph.D., Department of Cognitive Science, University of California, San Diego

3. Principal Investigator, Research Team, and Emergency Contact

In the case of emergencies, the Principal Investigator (Dr. Martin Zettersten) and the research team can be reached at 210-557-9525. The research team can also be reached at lillab@ucsd.edu. Please use the same phone number or email in case of emergencies.

4. Study Overview

This research study is being conducted to better understand language and cognitive development in infants, children, and adults.

We are inviting you to participate in a research study because you are in the right age range for the current study.

This form explains the research so that you may make an informed decision about participating.

- Research is voluntary - whether or not you participate is your decision. You can discuss your decision with others (such as family or friends).
- You can say yes, but change your mind later. If you say no, we will not hold your decision against you.
- Please ask the study team questions about anything that is not clear, and feel free to ask questions and mention concerns before, during, and after the research.
- You may consult with friends, family, or anyone else before deciding whether or not to participate in the study.
- You may download a copy of this consent form to keep.

The purpose of this study is to understand how individuals learn language and develop cognitive skills at different stages of development.

If you take part in this research, you will answer survey questions or play a few games or activities for between 5-60 minutes. You will have the opportunity to stop the study session at any time and you do not have to continue participating if you do not want to.

Participation in this research may involve some minimal risks or discomforts. There is a risk of potential loss of confidentiality, as well as the possibility of you experiencing boredom while completing these tasks. A more detailed description about risk can be found in Section 8. We aim to minimize the risk as much as possible.

There are no direct benefits to you from participating in this research. Some tasks are structured as games and may be enjoyable to complete. Your participation can contribute to scientific research, working towards better understanding language and cognitive development. You always have the option to not participate or withdraw from the study at any point in time. The alternative to being in this study is not to participate.

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More detailed information about this research study is provided below.

5. Whom can I talk to if I have questions?

If during your participation in the study you have questions or concerns, or if you think the research has hurt you, contact the research team at the numbers or emails listed in Section 3 on the first page of this form. You should not agree to participate in this study until the research team has answered any questions you have about the study, including information contained in this form.

If before or during your participation in the study you have questions about your rights as a research participant, or you want to talk to someone outside the research team, please contact:

- UC San Diego Office of IRB Administration at 858-246-4777 or irb@ucsd.edu

6. How many people will take part?

We plan to study around 4000 people, including children and adults across different types of studies.

7. What happens if you take part in the research?

As you read this form, please ask questions if something is not clear.

If you take part in this research, you will play a few games or activities on a computer or tablet for around 5-60 minutes. You may also answer some questions. Before each activity, you may watch a tutorial. These activities will usually involve listening to novel words together with novel images, and matching pictures with the corresponding words and concepts. Another example of a task includes responding to prompts and questions by drawing images, verbal answering or typing an answer. Other tasks will involve seeing various images and hearing speech, then making judgments about certain words and images.

This may take between 5-60 minutes (specified in the study description on the online platform). We may keep track of your spontaneous and natural behavior (e.g. what you are looking at) and your responses to any of the activities as part of the study. As part of our research, we may also collect audio recordings. As we are also interested in studying cognitive development, we may also collect eye tracking data via accessing your webcam. In these types of studies, you will be asked to grant access to both your camera and microphone of your device first. This access will terminate once the study has been completed. Please note that while audio may be recorded, there will be no video recordings stored. Any data collected may be used in research, as well as in other educational or research settings including professional meetings, publications, and public presentations to nonscientific groups. However, please note that your personal information won't be shared or collected, other than some general demographic information.

8. What are the risks and possible discomforts?

Our study involves minimal risks or discomforts. There is minimal risk of boredom from doing the tasks or anxiety about performance on the tasks. This risk will not be greater than any risk in daily life within the context of normal, daily life and interactions.

Risks of Loss of Confidential Information: There is a potential risk for the loss of confidentiality. To minimize this risk, we label all study information with subject codes, instead of your user ID or other information, which will be securely stored separately from any study information.

9. How will information about you be protected?

We will not be asking for personally identifying information and while we cannot guarantee complete confidentiality, we will limit access to information. These people who might have access might include:

- Members of the research team and other staff or representatives of UCSD whose work is related to the research or to protecting your rights and safety.

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- Representatives of Federal and other regulatory agencies who make sure the study is done properly and that your rights and safety are protected.

Any data or information obtained during your study session will be labeled with an anonymous participant ID, as opposed to your user ID. Any record linking identifying information, your user ID, and this ID will be kept securely and separately. Any and all audio recordings will be stored on our lab servers.

The results of this study may be published once the study is complete. In all findings, we will keep your data confidential. We may use these recordings in other educational and research settings to communicate our findings, such as academic conferences or classes.

We expect this study will be completed in 5 years. This is only an estimate and the actual time to complete the study may be longer or shorter depending on a number of factors.

10. Will I need to pay to participate in the research?

There will be no cost to you for your participation.

11. What if I agree to participate, but change my mind later?

You can stop participating at any time for any reason, and it will not be held against you or risk loss of compensation. If you stop participating, we may not be able to remove the information we have already collected about you because it is not linked to your identity.

12. What will happen to information collected from me?

The data we collect as part of the study may be used to answer other research questions or may be shared with other researchers for other projects. In the case of shared data, we will remove identifiers, and only deidentified data will be shared. Once we remove identifiers, we will not ask for your consent for the use or sharing of your data in other research. In addition, data that have been de-identified may be uploaded to a research data repository for other researchers to access and use.

While your privacy and confidentiality are very important to us and we will use safety measures to protect it, we cannot guarantee that your identity will never become known.

13. Will you be compensated for participating in this study?

To compensate for your time, you will receive compensation directly through an online survey platform (e.g., Amazon Mechanical Turk, Prolific, Connect), with a variable payment amount equivalent to \$12/hr.

This agreement for electronic consent applies only to your consent to participate in this research study.

- **I hereby give my consent to participate in this research study. I understand and agree to the terms as mentioned in the consent form.**

For a copy of this consent form, please download the PDF version of this consent form.